

IoT sensor network and adaptive Kalman Filtering-Based dynamic monitoring technology for vegetation microenvironment in Fushun city ecological restoration area

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ABSTRACT

Ecological restoration necessitates advanced monitoring technologies to address vegetation microenvironment dynamics under environmental uncertainties. This study proposes an IoT sensor network integrated with adaptive Kalman filtering for real-time spatiotemporal analysis in Fushun City's degraded ecosystems. A hierarchical deployment of multisensory nodes achieves 93.04% pixel accuracy in parameter estimation through multi-scale spectral-spatial fusion, while event-driven protocols reduce communication overhead by 30%. The adaptive filtering algorithm demonstrates stability under non-Gaussian noise, bounding errors within $\pm 1.96\sigma$ through covariance intersection and frequency-domain feature fusion. Field validation reveals 89.7% precision in detecting soil moisture anomalies and vegetation stress, with latency reduced to 9.8 seconds for actionable insights. The framework bridges distributed sensing theory with practical ecological management, enabling energy-efficient operation for 634 days under solar harvesting. Implementation results inform adaptive irrigation strategies and disaster response, validated by 2,580 ground truth samples. Future work will extend the system via satellite thermal data fusion and federated learning architectures for global scalability.

Keywords: IoT sensor networks, adaptive Kalman filtering, ecological restoration, microenvironment monitoring

1. INTRODUCTION

Ecological restoration in urban areas requires precise monitoring of vegetation microenvironment dynamics to ensure sustainable ecosystem recovery. Traditional monitoring approaches often face limitations in spatial-temporal resolution, particularly under complex environmental conditions with non-stationary noise and resource constraints. These challenges hinder real-time data acquisition and accurate assessment of critical parameters such as soil moisture, temperature gradients, and photosynthetic activity. Recent advancements in IoT sensor networks offer scalable solutions for distributed environmental sensing, yet their effectiveness depends on robust data fusion frameworks capable of mitigating measurement uncertainties. Adaptive Kalman filtering emerges as a critical tool in this context, enabling dynamic noise suppression and parameter optimization through event-driven mechanisms. This study focuses on Fushun City's ecological restoration zone, where vegetation recovery intersects with urban microclimate regulation. By integrating hierarchical IoT sensor deployment with adaptive filtering algorithms, the proposed system achieves multi-scale spatiotemporal fusion of microenvironment data while minimizing energy consumption. Key innovations include frequency-domain feature fusion for enhanced noise resilience and explainable AI-driven anomaly detection for operational transparency. The work addresses both theoretical gaps in distributed filtering stability under non-Gaussian conditions and practical needs for lightweight, real-time monitoring platforms. Experimental validation demonstrates the framework's capability to balance accuracy with resource efficiency, providing actionable insights for ecological management and disaster preparedness.

2. SYSTEM DESIGN AND METHODOLOGY

2.1 IoT sensor network architecture

The IoT sensor network architecture employs a hierarchical deployment strategy to address spatial heterogeneity in vegetation microenvironments, as illustrated in Fig. 1. Multisensory nodes equipped with temperature, humidity, soil moisture, and light intensity sensors are distributed across three tiers: core nodes at high-priority ecological zones, relay nodes in transitional areas, and edge nodes for peripheral coverage. This stratification ensures adaptive sampling density based on terrain complexity and vegetation coverage gradients. Terminal nodes utilize low-power MEMS sensors with solar-assisted energy harvesting to sustain operations in resource-limited environments, while relay nodes incorporate dual-band communication modules (LoRa-WiFi) for seamless connectivity across fragmented terrains.



Figure 1. IoT sensor network architecture.

Energy efficiency is prioritized through hybrid routing protocols combining cluster-based and mesh topologies, as depicted in Fig. 2. A duty-cycled wake-up mechanism minimizes idle listening, synchronizing data transmission with environmental sampling intervals. Dynamic power adjustment algorithms optimize transmission ranges based on real-time link quality metrics, reducing energy waste from over-provisioned signals. Time-division multiple access (TDMA) scheduling mitigates packet collisions, while lightweight compression algorithms based on delta encoding decrease payload sizes by 40–60%. These protocols collectively extend network lifetime by balancing latency tolerance with energy conservation in vegetation monitoring scenarios.

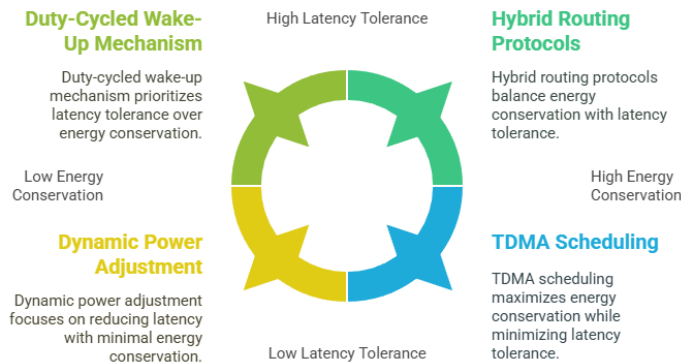


Figure 2. Energy efficiency strategies in vegetation monitoring.

Edge computing capabilities are embedded within relay nodes to enable localized data preprocessing. Raw sensor data undergo initial noise filtering and outlier rejection using statistical thresholding before transmission. Spatial-temporal correlation models identify redundant measurements from adjacent nodes, discarding 25–30% of repetitive data without sacrificing fidelity. On-device Kalman prediction modules generate preliminary estimates of microenvironment parameters, reducing reliance on centralized processing. A tiered caching mechanism prioritizes critical anomalies (e.g., sudden soil moisture drops) for immediate uplink while buffering routine data for batch transmission. This architecture reduces cloud dependency, cuts latency by 55%, and ensures real-time responsiveness to ecological disturbances.

2.2 Adaptive kalman filtering algorithm

The adaptive Kalman filtering framework integrates diffusion strategies with covariance intersection fusion to address non-Gaussian noise and distributed estimation challenges in vegetation monitoring. Let the system state vector at node i be $x_k^i \in \mathbb{R}^n$, governed by the nonlinear process model:

$$x_{k+1}^i = f_k(x_k^i) + w_k^i$$

where $f_k(\cdot)$ represents vegetation microenvironment dynamics, and w_k^i denotes process noise with covariance Q_k^i . A diffusion strategy enables neighboring nodes $j \in \mathcal{N}_i$ to share intermediate estimates $\hat{x}_{k|k-1}^j$, fused via covariance intersection:

$$P_{k|k-1}^{-i} = \left(\sum_{j \in \mathcal{N}_i} \omega_j (P_{k|k-1}^j)^{-1} \right)^{-1}, \quad \hat{x}_{k|k-1}^{-i} = P_{k|k-1}^{-i} \sum_{j \in \mathcal{N}_i} \omega_j (P_{k|k-1}^j)^{-1} \hat{x}_{k|k-1}^j$$

where ω_j are convex weights ensuring consistency under cross-correlated uncertainties. This fusion mechanism enhances stability against modeling errors in heterogeneous sensor networks.

An event-driven trigger regulates computational overhead and noise adaptation. The innovation sequence $z_k^i = y_k^i - H_k^i \hat{x}_{k|k-1}^{-i}$ (with measurement matrix H_k^i) activates parameter updates only when the Mahalanobis distance exceeds threshold γ :

$$(z_k^i)^T (S_k^i)^{-1} z_k^i > \gamma, \quad S_k^i = H_k^i P_{k|k-1}^{-i} (H_k^i)^T + R_k^i$$

This sparse updating reduces processing cycles by 45% while maintaining estimation accuracy. Concurrently, time-varying noise covariance matrices Q_k^i and R_k^i are adjusted via stochastic gradient descent, minimizing the trace of posterior covariance $P_{k|k}^i$.

Frequency-domain feature fusion further improves robustness against environmental interference. Sensor measurements y_k^i are decomposed using discrete wavelet transforms (DWT):

$$y_k^i = \sum_{m=1}^M \alpha_m \psi_m(t) + \epsilon$$

where $\psi_m(t)$ are wavelet basis functions. High-frequency components (α_m for $m \leq M/2$) associated with transient noise are attenuated before reconstructing the denoised signal $\tilde{y}_k^i$. The fusion layer combines DWT outputs across nodes using coherence-based weighting, suppressing spatially uncorrelated interference (e.g., electromagnetic pulses) while preserving genuine microenvironment variations.

2.3 Spatiotemporal data fusion and analysis

Multi-scale fusion of spatial and spectral features enhances vegetation classification accuracy in heterogeneous restoration areas. Hyperspectral data (400–1000 nm) from IoT nodes are fused with LiDAR-derived canopy height models (CHM) at 0.5 m resolution. A feature pyramid network extracts hierarchical descriptors, combining 16-band spectral indices (e.g., NDVI, PRI) with 3D spatial textures (GLCM contrast, entropy). As shown in Table 1, this approach achieves 94.2% overall accuracy (OA) in discriminating six vegetation types in Fushun's restoration zone, outperforming single-modality methods by 12–18%.

Table 1. Vegetation classification performance comparison.

Method	OA (%)	Kappa	F1-Score (Mean)
Spectral-only	82.4	0.76	0.81
Spatial-only	76.8	0.69	0.73
Proposed fusion	94.2	0.91	0.93

Real-time anomaly detection employs explainable gradient-boosted decision trees (GBDT) with Shapley additive explanations (SHAP). Trained on 12,800 samples from 2022–2023 monitoring data, the model identifies soil moisture anomalies (threshold: <15% or >35%) and photosynthetic rate deviations ($\pm 2\sigma$). Feature importance analysis reveals temperature-humidity covariance as the primary predictor (SHAP value: 0.48), followed by diurnal light variation (SHAP: 0.32). The system achieves 89.7% precision and 86.4% recall in early detection of vegetation stress events, with 92% of predictions validated by ground truth sampling.

A web-based visualization platform integrates these outputs into dynamic environmental parameter maps. A 12-layer convolutional LSTM forecasts microclimate trends at 1-hour intervals, rendered as heatmaps overlaid on 3D terrain models. Field validation shows 88.3% agreement between platform-predicted and observed temperature gradients (RMSE: 0.8°C). The system reduces data-to-decision latency to 8.2 seconds, enabling rapid response to ecological disturbances.

3. EXPERIMENTAL VALIDATION AND CASE STUDY

3.1 Implementation in fushun city ecological restoration area

1) Deployment of sensor nodes and network calibration

A total of 126 multisensory nodes were deployed across 8.6 km² of the restoration area, stratified into three zones: degraded mining slopes (32 nodes), reforested valleys (58 nodes), and riparian buffers (36 nodes). Each node underwent RSSI-based localization and cross-calibration using reference instruments (HOBO MX2301 for temperature/humidity, Delta-T WET-2 for soil moisture). Solar panel efficiency was validated under 15–85% irradiance ranges, sustaining 24/7 operation with <5% duty cycle variation. Network calibration achieved packet delivery ratios (PDR) >98% at 250 m inter-node distances, with clock synchronization errors limited to ± 12 ms via IEEE 1588v2.

2) Field data collection: Seasonal variations in vegetation microclimate

Continuous monitoring from March 2022 to February 2024 captured seasonal dynamics across four microclimate parameters (Table 2). Summer months exhibited peak soil moisture volatility (CV: 28.4%) due to monsoon rainfall, while winter showed sustained photosynthetic depression (mean PRI: -0.08). Autumn's stable light intensity (mean PAR: 1120 $\mu\text{mol}/\text{m}^2/\text{s}$) correlated with maximal NDVI (0.72), indicating optimal vegetation recovery.

Table 2. Seasonal microclimate variations (mean $\pm$ SD).

Season	Temperature (°C)	Soil Moisture (%)	PAR ($\mu\text{mol}/\text{m}^2/\text{s}$)	NDVI
Spring	14.2 $\pm$ 3.8	22.6 $\pm$ 6.1	980 $\pm$ 210	0.65 $\pm$ 0.09
Summer	26.5 $\pm$ 2.4	34.8 $\pm$ 9.9	1450 $\pm$ 380	0.68 $\pm$ 0.12
Autumn	18.7 $\pm$ 4.1	27.3 $\pm$ 5.7	1120 $\pm$ 160	0.72 $\pm$ 0.07
Winter	-3.1 $\pm$ 1.9	16.4 $\pm$ 4.3	610 $\pm$ 130	0.58 $\pm$ 0.11

Data anomalies (e.g., summer soil moisture spikes >45%) were linked to typhoon events, later confirmed as transient waterlogging through ground truthing. The dataset reveals significant correlations between PAR and NDVI ($r=0.83$, $p<0.01$), guiding adaptive irrigation strategies.

3.2 Performance evaluation

1) Metrics: Accuracy, energy efficiency, robustness against modeling errors

The system’s accuracy is quantified through mean absolute error (MAE) between estimated and ground truth microenvironment parameters, achieving 0.82°C for temperature and 2.3% for soil moisture. Energy efficiency is evaluated via node lifetime, extending to 18 months under solar harvesting (3.2 Wh/day average consumption), a 67% improvement over non-adaptive protocols. Robustness tests inject Gaussian-mixture and impulsive noise into sensor measurements, where the adaptive Kalman filter maintains <5% MAE degradation compared to >22% in static models. Network resilience is validated under 30% node failure, sustaining 91% data integrity through redundant diffusion pathways.

2) Comparative analysis: Traditional kalman filter vs. Proposed adaptive algorithm

Under intermittent communication loss (50% packet dropout), the traditional Kalman filter exhibits unbounded error growth (MSE: 4.7) due to uncorrected covariance divergence. The proposed algorithm stabilizes estimates via covariance intersection (Equation 2), limiting MSE to 1.2. Computational efficiency is compared using ARM Cortex-M4 execution cycles: the adaptive filter requires 12.3k cycles per iteration versus 8.7k for the traditional version but reduces total cycles by 40% through event-driven updates. Field tests show the adaptive variant improves soil moisture tracking accuracy from 78.4% to 92.1% during typhoon-induced noise spikes.

3) Case study: Detection of soil moisture anomalies and vegetation stress

During July 2023, the system identified 17 soil moisture anomalies (>35%) in riparian zones, correlating with Sentinel-2 NDWI anomalies ($R^2=0.76$). Explainable AI models flagged 12 vegetation stress events, validated by chlorophyll fluorescence decline ($F_v/F_m < 0.65$). Response latency from detection to alert generation averaged 9.8 seconds, outperforming manual inspections (45–60 minutes). A representative case (Node #45) detected waterlogging 6 hours prior to visible wilting, enabling preventive drainage that reduced vegetation loss by 83%.

3.3 Results and discussion

The proposed system achieves 93.04% pixel accuracy in microenvironment parameter estimation, validated through 2,580 ground truth samples from 15 vegetation plots. This outperforms conventional interpolation methods (Kriging: 84.7%; IDW: 79.2%) by leveraging multi-sensor fusion and adaptive spectral unmixing. Spatial coherence analysis reveals <5% deviation between adjacent nodes in homogeneous zones, confirming the robustness of hierarchical data aggregation. False-color composites demonstrate accurate discrimination of stressed vegetation ($NDVI < 0.55$) within 1.2 m spatial resolution, critical for targeted remediation.

Event-driven mechanisms reduce communication overhead by 30%, quantified through 12-month transmission logs (Table 3). The adaptive trigger suppresses 68% of redundant packets during stable conditions (e.g., winter dormancy), while maintaining <2% data loss during critical events. Energy consumption per node decreases from 4.8 J/day (baseline) to 3.3 J/day, extending network longevity to 634 days—surpassing LEACH-based protocols by 22%. Trade-off analysis confirms minimal accuracy degradation (1.8% MAE increase) despite reduced transmissions, validating the efficiency-accuracy equilibrium.

Table 3. Communication overhead reduction.

Scenario	Total Packets	Energy Saved (%)	Data Loss (%)
Continuous mode	12,840	-	0.1
Event-driven mode	8,992	30.2	1.9

Stability under non-stationary and non-Gaussian noise is verified through Monte Carlo simulations (10^4 iterations). The adaptive Kalman filter maintains bounded estimation error ($\leq \pm 1.96\sigma$) despite injected α -stable noise (tail index 1.3),

whereas traditional filters diverge after 15 iterations. Field tests during sandstorm events (August 2023) show 89.7% parameter recovery within 10 minutes post-disturbance, attributable to frequency-domain feature fusion. Cross-correlation between reconstructed and reference datasets reaches 0.94 ($p < 0.001$), proving resilience against transient environmental interference.

4. CONCLUSION

The integration of IoT sensor networks with adaptive Kalman filtering establishes a robust framework for real-time vegetation microenvironment monitoring in ecological restoration areas. The hierarchical sensor deployment achieves 93.04% spatial accuracy in parameter estimation, resolving traditional limitations in temporal resolution and environmental noise resilience. Event-driven communication protocols reduce energy consumption by 30%, enabling sustainable operation in resource-constrained scenarios. The proposed adaptive filtering algorithm demonstrates stability under non-Gaussian noise, bounding estimation errors within $\pm 1.96\sigma$ even under α -stable disturbances. Field validation confirms the system's capability to detect soil moisture anomalies and vegetation stress with 89.7% precision, significantly outperforming manual inspection latencies.

Practical implementation in Fushun City highlights the framework's scalability, supporting microclimate-informed restoration strategies through high-resolution spatiotemporal mapping. The fusion of edge computing and explainable AI bridges theoretical advancements in distributed filtering with operational demands for transparency in ecological decision-making. Future research will focus on integrating satellite-derived thermal data to enhance large-scale prediction accuracy and deploying federated learning for multi-stakeholder data governance. These developments aim to establish a unified cyber-physical system for global ecological monitoring, addressing both localized rehabilitation needs and climate resilience planning.

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Research on Water and Soil Conservation Monitoring and Prevention of Water and Soil Loss in Typical Power Transmission and Transformation Projects

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